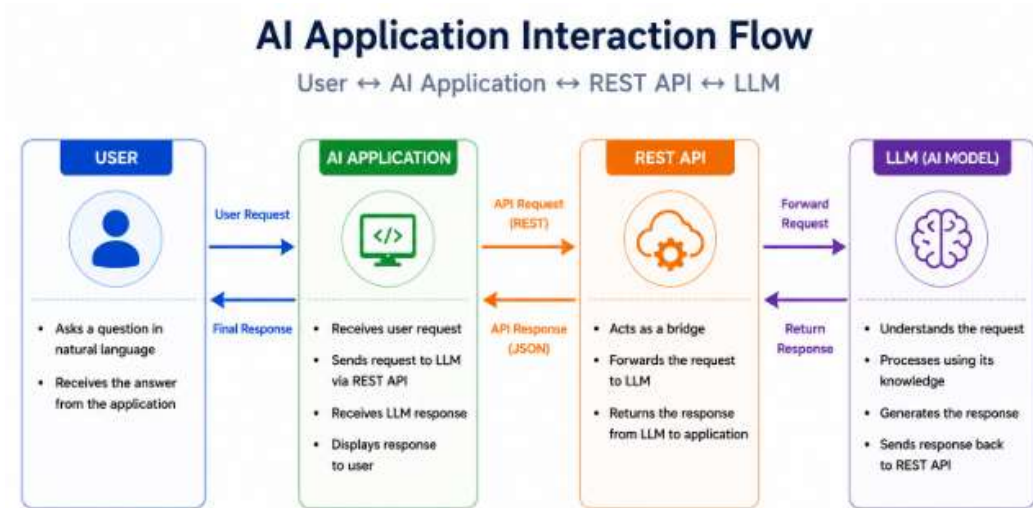


How to Use Existing LLMs to Build AI Applications?

- As AI Application Developers, we don't build AI models —we use existing pre-trained AI Models to build AI Applications.
- Most AI models used in AI applications are accessed through REST APIs.
- Let's build our first AI-powered application: AI-Powered Arithmetic Calculator.
- Our application should understand natural language and generate the correct answer.
- **Example:** "What is the addition of 2 and 4?" -> Return Answer as 6
- **How do we add AI capabilities to our application?**
 - Using AI Models
 - Use AI Model as LLM for AI-Powered Arithmetic Calculator because it has to work on text
- **How do we use an LLM from our application?**
 - AI applications communicate with LLMs through REST APIs.



Where Does the LLM Run ?

- **We used REST APIs to access LLMs but if you see our code we used two types of REST APIs to access LLMS**

- <https://api.groq.com/openai/v1/chat/completions>
- <http://localhost:11434/api/generate>
- **Note:** The main difference is where the LLM is running.

- **Based on where the LLM runs, they are classified into two types**

- **Hosted LLMs** : where the LLMs run on cloud servers provided by AI companies
- **Local LLMs** : where the LLMs run on our own machine or server.

- **Hosted LLMs:-** can be accessed in two ways

- **Direct AI Providers**

- OpenAI
- Google
- Anthropic
- Groq
- Mistral AI
- Cohere

- **Cloud AI Platforms**

- AWS Bedrock
- OCI Generative AI
- Azure OpenAI Service
- Google Vertex AI

- **Local LLMs (Self-hosted):**

- Run the LLMs on your own computer or server.
- Some of the LLMs to run locally :
 - Llama
 - DeepSeek
 - Mistral
 - Falcon

- **Note:** We need Runtimes like Ollama, LM Studio to run Open LLMs on our computer or server

Notes:

- ❖ Local LLMs: No API usage charges, but you need sufficient hardware to run the models.
- ❖ Hosted LLMs: The provider manages the infrastructure for you. You typically pay only for what you use, and pricing is generally based on the number of input and output tokens processed.